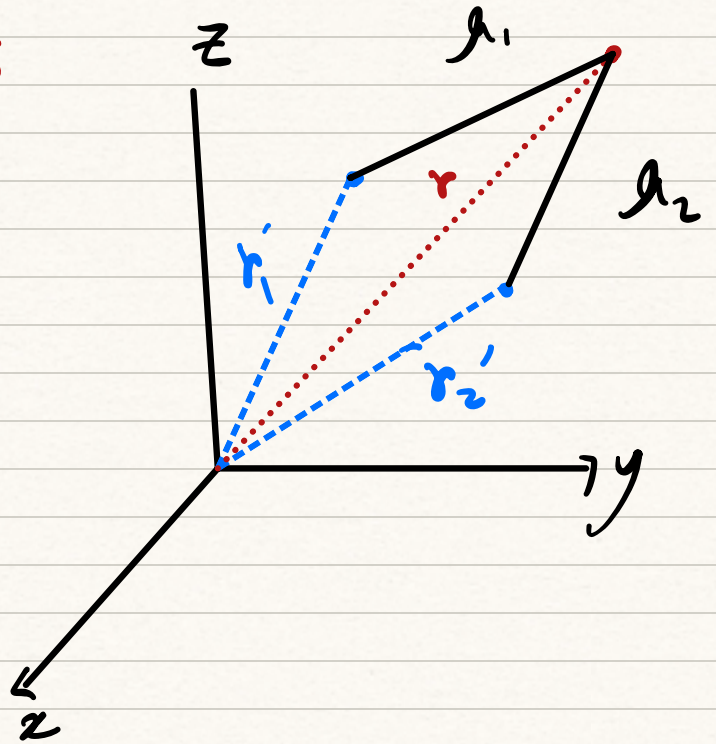


# Chap2: Electrostatics

## Sec1 | Coulomb's Law

### 1) Discrete Charges

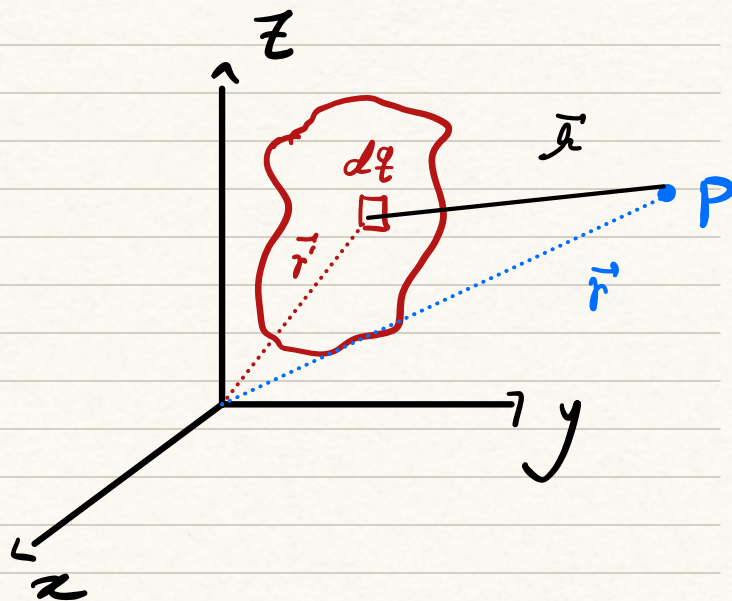
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$



## 2) Continuous charged objects:

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \hat{r}$$

$$E = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2}$$



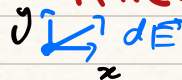
• where  $dq = \rho d\tau$

$$\Rightarrow Q = \iiint \rho d\tau$$

### • The Procedure to find $\vec{E}$ :

i) Consider the components of  $d\vec{E}$ , i.e.  $d\vec{E} = dE \cos\theta + dE \sin\theta$

ii) use the symmetry, if it is applicable, to cancel one components i.e.

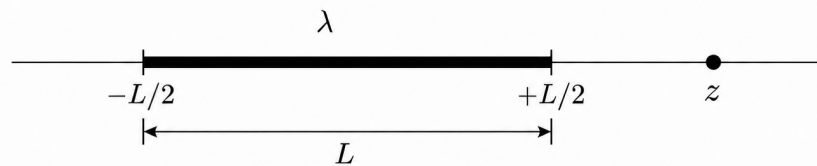


iii) Find  $\mu$ ,  $dq$  in term of integration variable

iv) Express  $\sin\theta$ ,  $\cos\theta$  in terms of  $\mu$ .

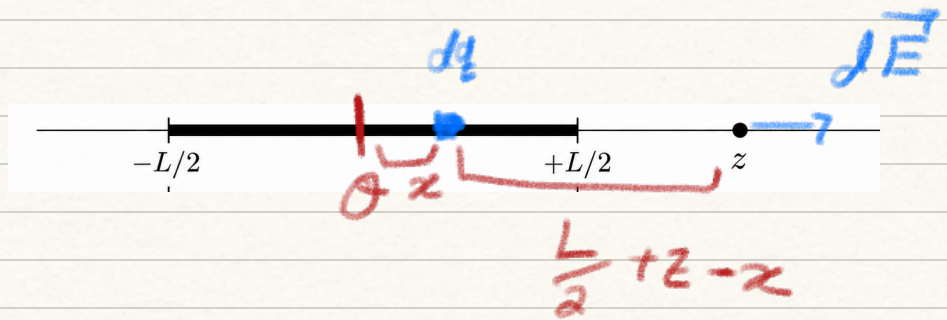
v) integrate  $E = \int dE$

ExP 1.1) A uniform charged wire of length  $L$ , & charge density  $\lambda$  electric field at point  $P$ .



$$dq = \lambda dx$$

$$r = \frac{L}{2} + z - x$$



$$dE = \frac{\lambda dx}{4\pi\epsilon_0 \left(\frac{L}{2} + z - x\right)^2}$$

$$E = \frac{\lambda}{4\pi\epsilon_0} \int_{-L/2}^{+L/2} \frac{dx}{\left(\frac{L}{2} + z - x\right)^2}$$

$$= \frac{\lambda}{4\pi\epsilon_0} (-1) \left(\frac{L}{2} + z - x\right)^{-1} \Big|_{-L/2}^{+L/2}$$

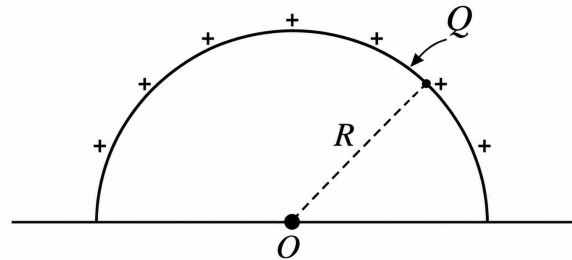
$$= \frac{\lambda}{4\pi\epsilon_0} \left[ \frac{1}{L+z} - \frac{1}{z} \right]$$

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \left[ \frac{1}{L+z} - \frac{1}{z} \right] \hat{x}$$

## Exp 1.2)

Consider a uniformly charged semicircular arc of radius  $R$  and total charge  $Q$ .

Find  $\vec{E}$  at Point  $O$

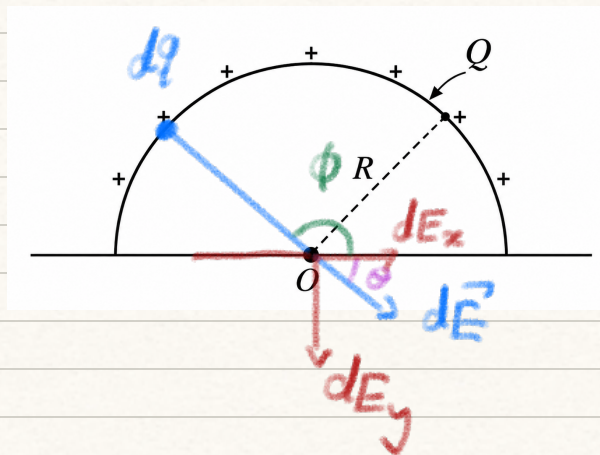


• By symmetry,  $E_x = 0$

$$dE_y = -dE \sin \theta$$

$$dq = \lambda dl = \lambda R d\phi$$

$$\lambda = \frac{Q}{\frac{1}{2} \times 2\pi R} = \frac{Q}{\pi R}$$



$$\Rightarrow dq = \frac{Q}{\pi} d\phi$$

$$a = R$$

$$\Rightarrow dE_y = -dE \sin \phi ; \phi + \theta = 180^\circ$$

$$\Rightarrow dE_y = \frac{-Q d\phi}{4\pi\epsilon_0 \pi R^2} \sin \phi$$

$$E_y = \frac{-Q}{4\pi\epsilon_0 \pi R^2} \int_0^\pi \sin\phi \, d\phi$$

$$= \frac{-2Q}{4\pi\epsilon_0 \pi R^2}$$

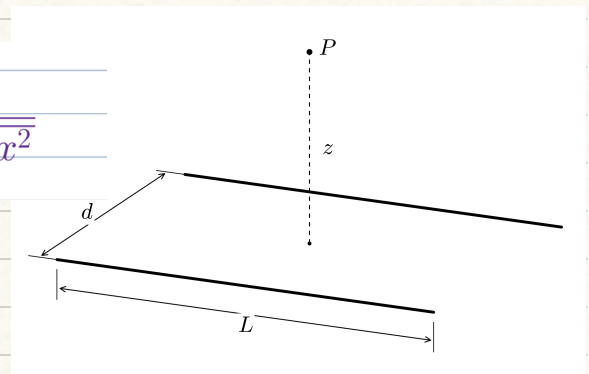
$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2Q}{\pi R^2} (-\hat{y})$$

### Exp 1.3

Consider two uniformly charged parallel wires of length  $L$  and charge density  $\lambda$ , separated by  $d$ .

Find  $\vec{E}$  at Point  $P$  which lies between them at height  $z$ .

Hint:  $\int \frac{dx}{(\alpha^2 + x^2)^{3/2}} = \frac{1}{\alpha^2} \frac{x}{\sqrt{\alpha^2 + x^2}}$



• By symmetry  $E_x = E_y = 0$

$$dE_z = 2 dE \cos \theta$$

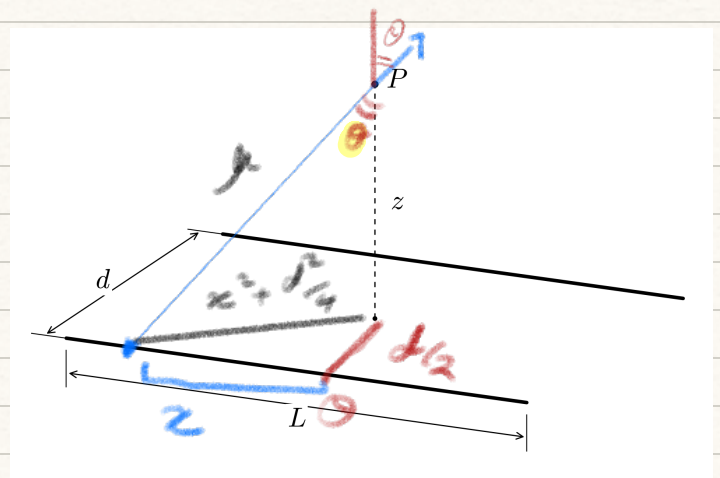
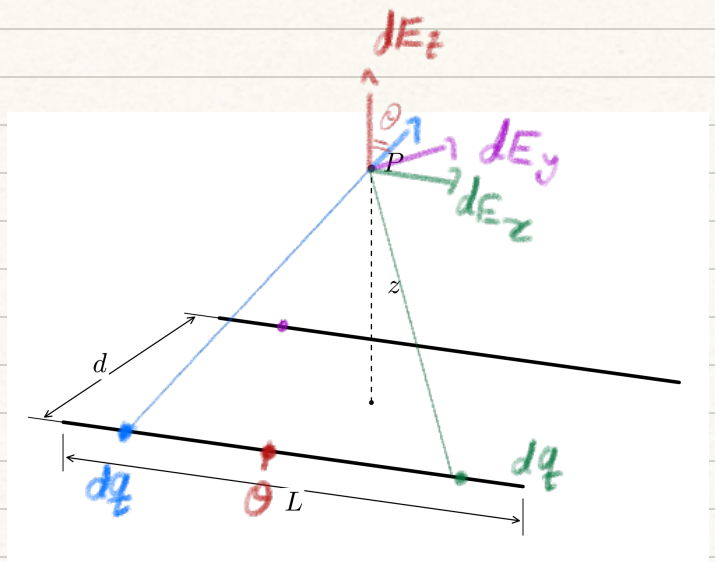
$$r^2 = z^2 + \frac{d^2}{4} + x^2$$

$$dq = \lambda dx$$

$$\cos \theta = \frac{z}{r}$$

so

$$dE_z = \frac{2}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \cdot \frac{z}{r}$$



$$= \frac{2}{4\pi\epsilon_0} \frac{z\lambda dx}{\left(x^2 + z^2 + \frac{d^2}{4}\right)^{3/2}}$$

$$E_z = \frac{z\lambda}{2\pi\epsilon_0} \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{dx}{\left(x^2 + z^2 + \frac{d^2}{4}\right)^{3/2}}$$

$$= \frac{z\lambda}{\pi\epsilon_0} \int_0^{\frac{L}{2}} \frac{dx}{\left(x^2 + z^2 + \frac{d^2}{4}\right)^{3/2}}$$

$\underbrace{\hspace{1.5cm}}_{\approx L}$

$$= \frac{z\lambda}{\pi\epsilon_0} \cdot \frac{1}{\left(z^2 + \frac{d^2}{4}\right)} \cdot \left[ \frac{x}{\sqrt{x^2 + z^2 + \frac{d^2}{4}}} \right]_0^{\frac{L}{2}}$$

$$= \frac{\lambda L z}{2\pi\epsilon_0 \left(z^2 + \frac{d^2}{4}\right) \sqrt{\frac{L^2}{4} + z^2 + \frac{d^2}{4}}}$$

Notice at large  $z$ ,

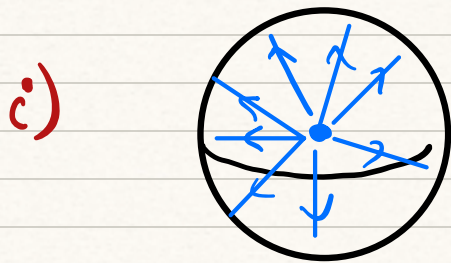
$$E \approx \frac{\lambda L z}{2\pi\epsilon_0(z^2) \cdot z} = \frac{\lambda L}{2\pi\epsilon_0 z^2}$$

$$= \frac{2\lambda L}{4\pi\epsilon_0 z^2} = \frac{Q_{tot}}{4\pi\epsilon_0 z^2}$$

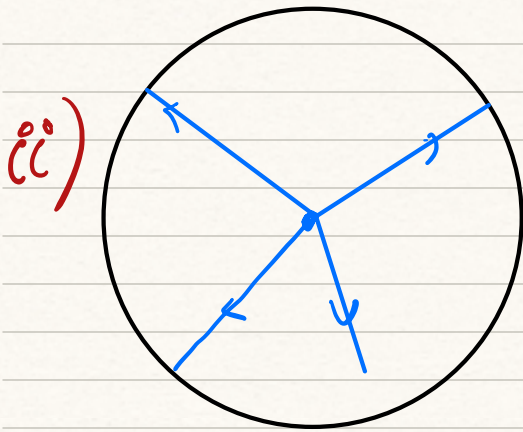
## Sec 2 | Gauß' law

### • Intuition:

Consider a charge  $q$  which is enclosed by two surfaces:



- many  $\vec{E}$  lines
- small area



- a few  $\vec{E}$  lines
- large area

• consider the electric flux

$$\Phi_E = \int \vec{E} \cdot d\vec{a}$$

notice  $d\vec{a} \sim r^2$  and  $E \sim \frac{q}{r^2}$

$$\Rightarrow \Phi_E \sim \frac{q}{r^2} \cdot r^2 \sim q$$

Conclusion:  $\Phi_E$  over closed surface should depend only on charges inside!

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Where  $Q_{\text{enc}} = \int_V \rho \, d\tau$

• Gaussian Surfaces:

1- Spherical:  $d\vec{a} = r^2 \sin\theta \, d\theta \, d\phi \, \hat{r}$

2- Cylindrical:  $d\vec{a} = s \, d\theta \, d\phi \, \hat{s}$

3- Pillbox:  $da = dx \, dy \, \hat{z}$

Ex 2.1) Consider an insulating sphere of radius  $R$  and charge density  $\rho(r) = \rho_0 \left(1 - \frac{4r}{3R}\right)$ .

Find  $E$  inside and outside.

• Inside

$$d\vec{a} = r^2 \sin\theta d\theta d\phi \hat{r}$$

$$\iint \vec{E} \cdot d\vec{a}$$

$$= \int_0^{2\pi} \int_0^\pi E r^2 \sin\theta d\theta d\phi$$

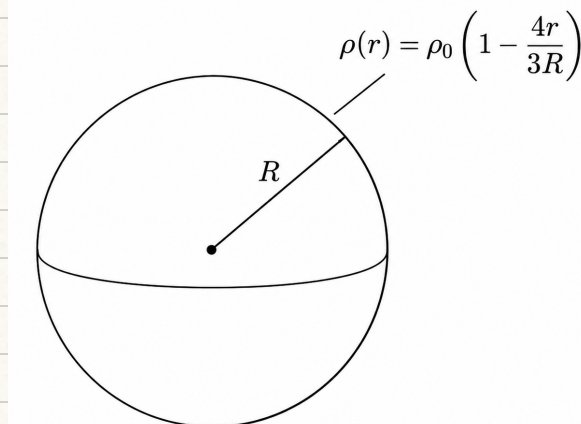
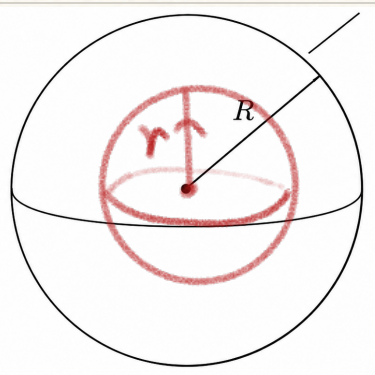
$$= E r^2 (4\pi)$$

\* using  $\int_0^{2\pi} \int_0^\pi \sin\theta d\theta d\phi = 4\pi$

$$Q_{enc} = \iiint \rho r^2 \sin\theta d\theta d\phi dr = \int_0^r \int_0^{2\pi} \int_0^\pi \rho_0 \left(r^2 - \frac{4r^3}{3R}\right) \sin\theta d\theta d\phi dr$$

$$= 4\pi \rho_0 \int_0^r \left(r^2 - \frac{4r^3}{3R}\right) dr = \frac{4\pi \rho_0}{3} \left[ r^3 - \frac{r^4}{R} \right]$$

$$= \frac{4\pi \rho_0}{3} r^3 \left[ 1 - \frac{r}{R} \right]$$



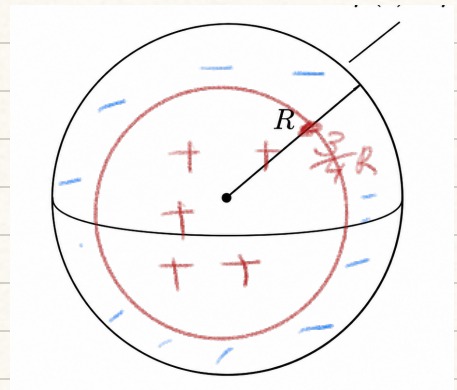
$$\text{So } E r^2 (4\pi) = \frac{4\pi \rho_0}{3\epsilon_0} r^3 \left[ 1 - \frac{r}{R} \right]$$

$$\vec{E}_{\text{in}} = \frac{\rho_0 r}{3\epsilon_0} \left[ 1 - \frac{r}{R} \right] \hat{r}$$

• Outside:  $r > R$

$$Q_{\text{enc}} = 0 \Rightarrow$$

$$E_{\text{out}} = 0$$



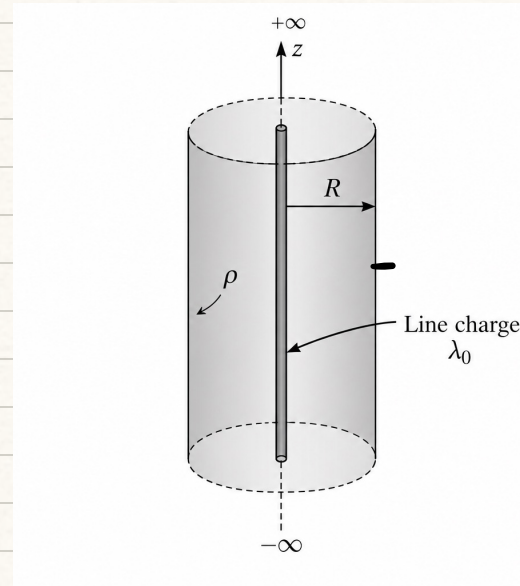
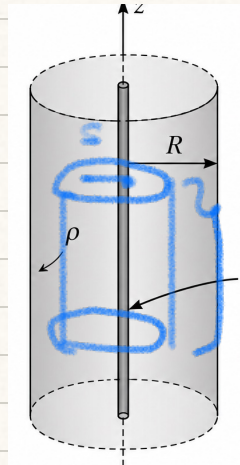
Ex 2.2) An infinite insulating cylinder has radius  $R$  and charge density  $\rho(s) = \frac{\rho_0 s}{R}$ . A wire runs along its axis with linear charge density  $-\lambda_0$ .

Find  $\vec{E}$  inside & outside.

• Inside:

$$d\vec{a} = s dz d\phi \hat{s}$$

$$\iint \vec{E} \cdot d\vec{a}$$



$$= \int_0^{2\pi} \int_0^l E s dz d\phi = E(2\pi l s)$$

$$Q_{enc} = -\lambda_0 l + \iiint \rho s ds d\phi dz$$

$$= -\lambda_0 l + 2\pi l \int_0^s \frac{\rho_0 s}{R} s ds$$

$$= -\lambda_0 l + \frac{2\pi l \rho_0}{3R} s^3$$

$\Rightarrow$  Gauss's law

$$E(2\pi s l) = \frac{1}{\epsilon_0} \left( \frac{2\pi l \rho_0}{3R} s^3 - \lambda_0 l \right)$$

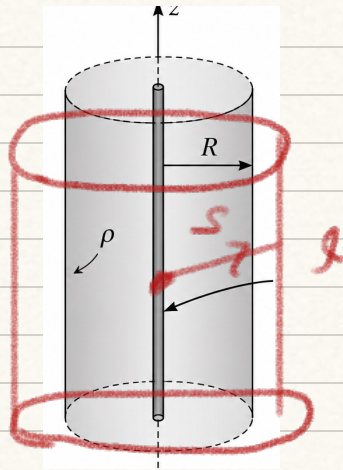
$\Rightarrow$

$$\vec{E}_{in} = \left( \frac{\rho_0}{3\epsilon_0 R} s^2 - \frac{\lambda_0}{2\pi\epsilon_0 s} \right) \hat{s}$$

• outside

$$\iint \vec{E} \cdot d\vec{a}$$

$$= E(2\pi s l)$$



$$Q_{enc} = -\lambda_0 l + \frac{\rho_0}{R} \cdot 2\pi \cdot l \int_0^R s^2 ds$$

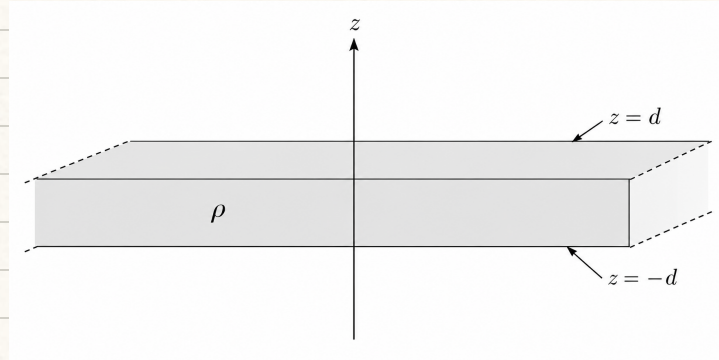
$$= -\lambda_0 l + \frac{2\pi \rho_0 l}{3} R^2$$

$$\Rightarrow \vec{E}_{\text{out}} = \left( \frac{\rho_0 R^2}{3\epsilon_0} \frac{1}{s} - \frac{\lambda_0}{2\pi\epsilon_0 s} \right) \hat{s}$$

Ex P 2.3) An infinite insulating slab occupies

$-d \leq z \leq d$  with charge density  $\rho(z) = \rho_0 \left(1 - \frac{z^2}{d^2}\right)$

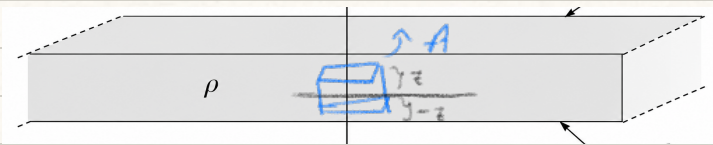
Find  $\vec{E}$



• Inside:  $|z| < d$

$$d\vec{a} = dx dy \hat{z} \text{ (at } z)$$

$$d\vec{a} = dx dy (-\hat{z}) \text{ (at } -z)$$



$$\iint \vec{E} \cdot d\vec{a} = 2E \iint dx dy = 2EA$$

$$Q_{\text{enc}} = \iiint \rho_0 \left(1 - \frac{z^2}{d^2}\right) dz dx dy$$

$$= A \rho_0 \int_{-z}^z \left(1 - \frac{z^2}{d^2}\right) dz = 2A \rho_0 \left(z - \frac{z^3}{3d^2}\right)$$

Gauss's law:

$$2EA = \frac{1}{\epsilon_0} \left( 2A \rho_0 \left( z - \frac{z^3}{3d^2} \right) \right)$$

$$\vec{E}_{in} = \frac{\rho_0}{\epsilon_0} \left( z - \frac{z^3}{3d^2} \right) \hat{z}$$


---

• Outside:  $|z| > d$

$$Q_{enc} = 2A\rho_0 \left[ d - \frac{d^3}{3d^2} \right] = 2A\rho_0 \left[ d - \frac{d}{3} \right]$$

$$= \frac{4}{3} A\rho_0 d$$

by Gauss's law:

$$2EA = \frac{4}{3} A\rho_0 d \Rightarrow E = \frac{2}{3} \rho_0 d$$

$$\vec{E}_{out} = \begin{cases} \frac{2}{3} \rho_0 d (-\hat{z}) ; & z < -d \\ \frac{2}{3} \rho_0 d \hat{z} ; & z > d \end{cases}$$

## Sec 3 | Electric Potential

### • Mathematically

we have  $\nabla \times \vec{E} = 0$

$$\Rightarrow \vec{E} = -\nabla V$$

for some  $V$  scalar function.

### • Physical meaning:

work done by  $\vec{E}$  on a point  $Q$

$$W = \int_a^b \vec{F} \cdot d\vec{r} = Q \int_a^b \vec{E} \cdot d\vec{r} = Q [V(a) - V(b)]$$

$$\Rightarrow V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

• if consider a point  $\theta$ ,  $V(\theta) = 0$

$$V(r) \equiv - \int_{\theta}^r \vec{E} \cdot d\vec{l}$$

Notice,

Given  $V$ , we have  $\vec{E}$

$$\vec{E} = -\nabla V$$

& Given  $\vec{E}$ , we have  $V$

$$V(r) = -\int_a^r \vec{E} \cdot d\vec{l}$$

• Poisson's Equation & reverse.

Given  $V$

$$\nabla^2 V = \frac{-\rho}{\epsilon_0} \quad \text{Poisson's}$$

How about the opposite?

Given  $\rho$ ;

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau'$$

• To Find  $V$ , either

a) Find  $\vec{E}$ , then

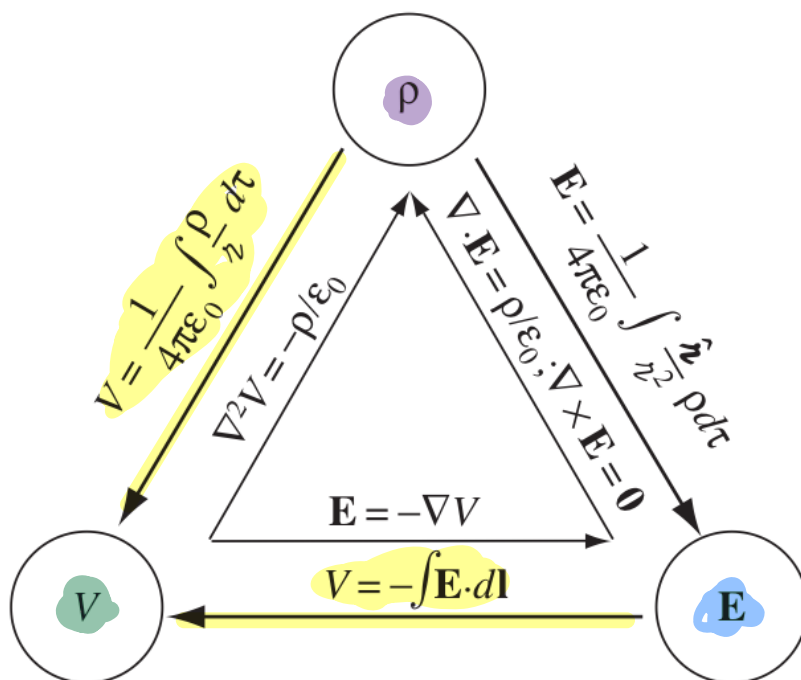
$$V = - \int \vec{E} \cdot d\vec{l}$$

or

b) Given  $\rho$ , find  $V$

$$V = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r} d\tau'$$

# The Electromagnetic Trinity



### Exp 3.1

given  $\sigma(r) = \frac{k}{r}$ , find  $V$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1} \left( \frac{x}{a} \right)$$

since  $\sigma$  is given, we need to find  $h$ .

$$h^2 = s^2 + z^2$$

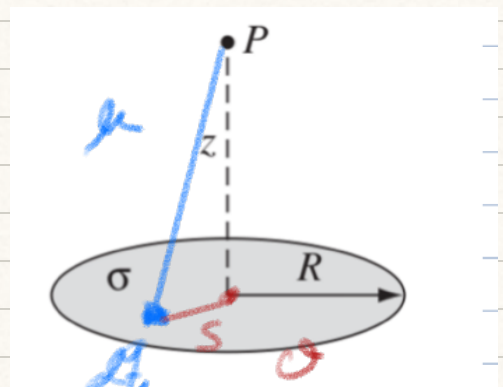
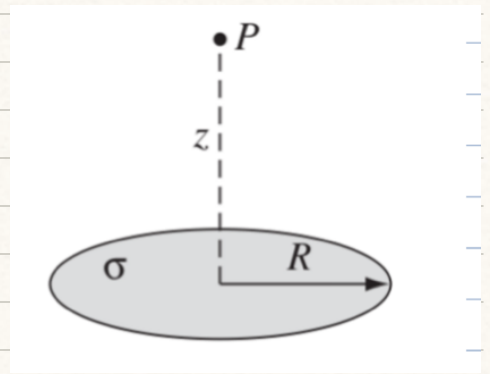
$$h = \sqrt{s^2 + z^2}$$

$$V = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma(s)}{\sqrt{s^2 + z^2}} s ds d\phi$$

$$= \frac{2\pi}{4\pi\epsilon_0} \int_0^R \frac{k}{\sqrt{s^2 + z^2}} ds$$

$$= \frac{k}{2\epsilon_0} \sinh^{-1} \left( \frac{s}{z} \right) \Big|_{s=0}^{s=R}$$

$$\Rightarrow V = \frac{k}{2\epsilon_0} \sinh^{-1} \left( \frac{R}{z} \right)$$



## Exp 3.2 Two conductive concentric spherical shells

Find  $V(r)$

Find  $\vec{E}$  for  $a < r < b$ .

by Gauss's law

$$E(4\pi r^2) = \frac{1}{\epsilon_0} Q_{\text{enc}} = \frac{q}{\epsilon_0}$$

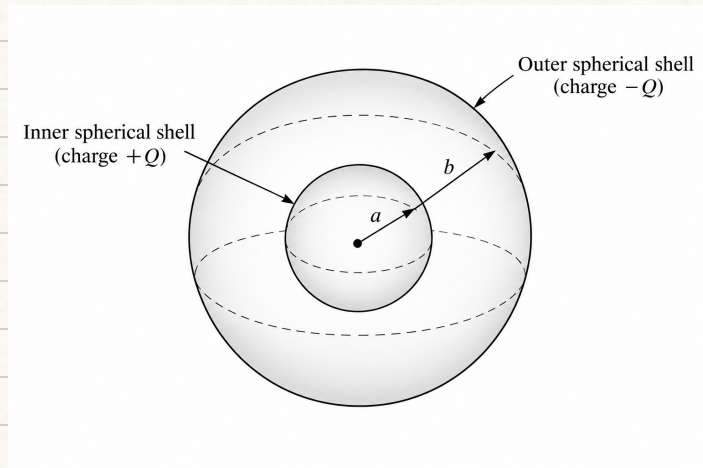
$$\Rightarrow \vec{E} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r}$$

and  $\vec{E} = 0$  for  $r > b$  &  $r < a$

$$V(r) = - \int_{\infty}^r \vec{E} \cdot d\vec{r} = - \int_{\infty}^r E dr$$

• for  $r > b$

$$V(r) = \int_{\infty}^r \cancel{E} dr = 0$$



• for  $a \leq r \leq b$

$$V(r) = - \int_{\infty}^r E dr - \int_b^r E dr$$

$$= - \int_b^r \frac{q}{4\pi\epsilon_0 r^2} dr = \frac{q}{4\pi\epsilon_0} \left( \frac{1}{r} - \frac{1}{b} \right)$$

• for  $r \leq a$ :

$$V(r) = - \int_{\infty}^b E dr - \int_b^a \frac{q}{4\pi\epsilon_0 r^2} dr$$

$$= \frac{Q}{4\pi\epsilon_0} \left( \frac{1}{a} - \frac{1}{b} \right)$$

$$V(r) = \begin{cases} 0 & ; \quad r > b \\ \frac{q}{4\pi\epsilon_0} \left( \frac{1}{r} - \frac{1}{b} \right) & ; \quad a \leq r \leq b \\ \frac{q}{4\pi\epsilon_0} \left( \frac{1}{a} - \frac{1}{b} \right) & ; \quad r \leq a \end{cases}$$

as  $E = 0$  in both  $r > b$  &  $r \leq b$

$V$  is fixed.

# Sec 4 | Energy & Capacitance

• Electrostatic energy:

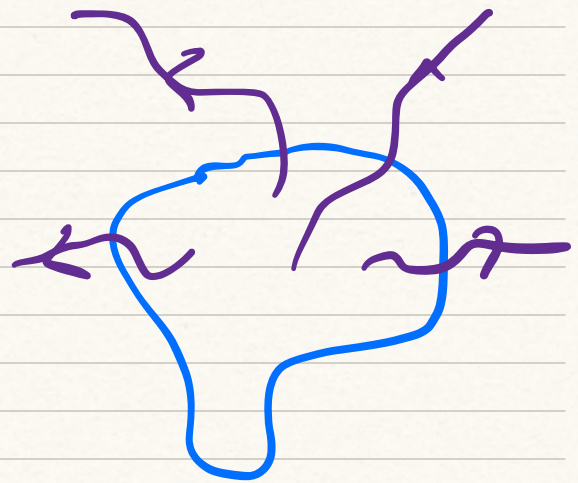
$$W = \frac{1}{2} \int \rho \, d\tau$$

over the charge distribution only ☹️



$$W = \frac{1}{2} \epsilon_0 \int E^2 \, d\tau$$

over all space!



\* The charge distribution is **localized**  
where its disturbance to E field is not.

- Capacitance:

$$C \equiv \frac{Q}{V} \Rightarrow V = \frac{Q}{C}$$

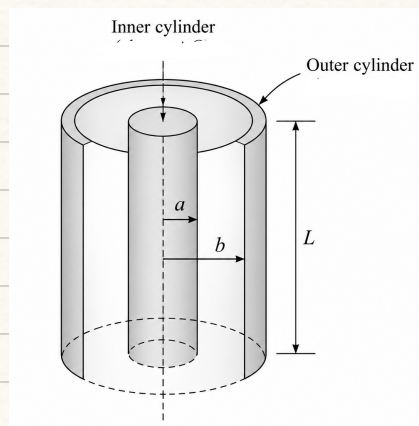
\* to find  $C$ , find  $V$  in terms of  $Q$

i.e.  $V = (\dots) Q$

$$\Rightarrow C = \frac{1}{(\dots)}$$

- $Q$  is not important, so we can assume it.

Exp 4.1 Two very long coaxial cylinders of radii  $a < b$ . With inner carries  $+\lambda$  & outer  $-\lambda$



a) Find  $V(a) - V(b)$

b) Find Capacitance Per length

c) Find the energy  $W$  by two methods.

Panta:

for  $a \leq s \leq b$ , by Gauss's law

$$E(2\pi Ls) = \frac{Q_{enc}}{\epsilon_0} = \frac{1}{\epsilon_0} (\lambda L)$$

$$\Rightarrow \vec{E} = \frac{\lambda}{2\pi\epsilon_0 s} \hat{s}$$

So

$$V(a) - V(b) = - \int_b^a \frac{\lambda}{2\pi\epsilon_0 s} ds$$

$$= \frac{\lambda}{2\pi\epsilon_0} \ln s \Big|_a^b = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

• Part b:

we got  $\Delta V = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$

$$\Rightarrow \Delta V = \frac{1}{L} \frac{L\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

but  $L\lambda = Q$

$$\Rightarrow \Delta V = \frac{1}{L} \frac{Q}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

$$\Rightarrow C = \frac{2\pi\epsilon_0 L}{\ln\left(\frac{b}{a}\right)}$$

$$\Rightarrow \boxed{\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln\left(\frac{b}{a}\right)}}$$

Part C:

Method I:

$$W = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau$$

$$= \frac{\epsilon_0}{2} \left[ \int_0^a E^2 d\tau + \int_a^b E^2 d\tau + \int_b^\infty E^2 d\tau \right]$$

$$= \frac{\epsilon_0}{2} \int_0^L \int_0^{2\pi b} \int_0^a \left( \frac{\lambda}{2\pi\epsilon_0} \right)^2 \frac{1}{s^2} \cdot s \, ds \, d\phi \, dz$$

$$= \frac{\epsilon_0}{2} \frac{\lambda^2}{4\pi^2\epsilon_0^2} 2\pi L \cdot \ln\left(\frac{b}{a}\right)$$

$$= \frac{\lambda^2 L}{4\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$

Method II:

$$W = \frac{1}{2} \iint \lambda V$$

$$= \frac{1}{2} \left[ V_a \iint \lambda da - V_b \iint \lambda da \right]$$

$$= \frac{1}{2} \left[ V_a \lambda L - V_b \lambda L \right]$$

$$= \frac{1}{2} \lambda L [V_a - V_b]$$

$$= \frac{1}{2} \lambda L \left[ \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{b}{a}\right) \right]$$

$$= \frac{\lambda^2 L}{4\pi\epsilon_0} \ln\left(\frac{b}{a}\right)$$